

Complex Numbers

ANSWER KEY



Powers of i

- 1 Evaluate each power of i , using $i^2 = -1$ and the fact that powers of i repeat every 4:

a $i^3 = -i$

b $i^4 = 1$

c $i^7 = -i$

d $i^{22} = -1$

- 2 Evaluate i^{101} .

$$101 = 4(25) + 1, \text{ so } i^{101} = (i^4)^{25} \cdot i^1 = 1 \cdot i = i.$$

Addition and subtraction

- 3 Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

$$= 1 + 9i$$

- 4 Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

$$= 1 - 7i$$

- 5 Find $(7 + 2i) - (3 - 5i)$.

$$= 4 + 7i$$

- 6 Find $(4 - i) - (-2 + 6i)$.

$$= 6 - 7i$$

Multiplication

- 7 Multiply $(2 + 3i)(1 - 4i)$.

$$(2)(1) - (3)(-4) + [(2)(-4) + (3)(1)]i = 14 - 5i$$

- 8 Multiply $(-1 + 2i)(3 + i)$.

$$(-1)(3) - (2)(1) + [(-1)(1) + (2)(3)]i = -5 + 5i$$

- 9 Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.

$$= 25 - (-4) + 0i = 29. \text{ It is a product of conjugates } z\bar{z} = |z|^2 \text{ (Theorem 1.1), which is always real.}$$

Conjugate and modulus

- 10 For $z = 6 - 8i$, find $\bar{z}$ and $|z|$.

$$\bar{z} = 6 + 8i, \text{ and } |z| = \sqrt{6^2 + 8^2} = \sqrt{100} = 10.$$

- 11 For $z = -5 + 12i$, find $\bar{z}$ and $|z|$.

$$\bar{z} = -5 - 12i, \text{ and } |z| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

- 12 Show that $z + \bar{z}$ is always real for $z = a + bi$.

$$z + \bar{z} = (a + bi) + (a - bi) = 2a, \text{ which is real since } a \text{ is real.}$$

Division

- 13 Simplify $\frac{3 + 4i}{1 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

$$\text{Multiply by } \frac{1 + 2i}{1 + 2i}: \text{ numerator} = (3)(1) - (4)(2) + [(3)(2) + (4)(1)]i = -5 + 10i; \text{ denominator} = 1^2 + 2^2 = 5. \text{ So the quotient is } -1 + 2i.$$

- 14 Simplify $\frac{1 + i}{1 - i}$.

$$\text{Multiply by } \frac{1 + i}{1 + i}: \text{ numerator} = (1)(1) - (1)(1) + [(1)(1) + (1)(1)]i = 0 + 2i; \text{ denominator} = 1^2 + 1^2 = 2. \text{ So the quotient is } i.$$

- 15 Simplify $\frac{7 - i}{3 + 2i}$.

$$\text{Multiply by } \frac{3 - 2i}{3 - 2i}: \text{ numerator} = (7)(3) - (-1)(-2) + [(7)(-2) + (-1)(3)]i = 19 - 17i; \text{ denominator} = 3^2 + 2^2 = 13. \text{ So the quotient is } \frac{19}{13} - \frac{17}{13}i.$$

Quadratic equations with complex roots

- 16 Solve $x^2 + 9 = 0$.

$$x^2 = -9 \Rightarrow x = \pm 3i.$$

- 17 Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

$$\text{Discriminant} = 36 - 52 = -16, \text{ so } \sqrt{-16} = 4i. x = \frac{6 \pm 4i}{2} = 3 \pm 2i.$$

- 18 Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.

$$\text{Discriminant} = 4 - 40 = -36, \text{ so } \sqrt{-36} = 6i. x = \frac{-2 \pm 6i}{4} = -\frac{1}{2} \pm \frac{3}{2}i.$$